

TRANSFORMED • STRATEGY RECONCILIATION

Plan vs Pipeline

Eight domains where the *AI Learning Product* strategy doc either extends, duplicates, or collides with what's already built. Grounded in the actual pipeline code, not the tracker's claims about it.

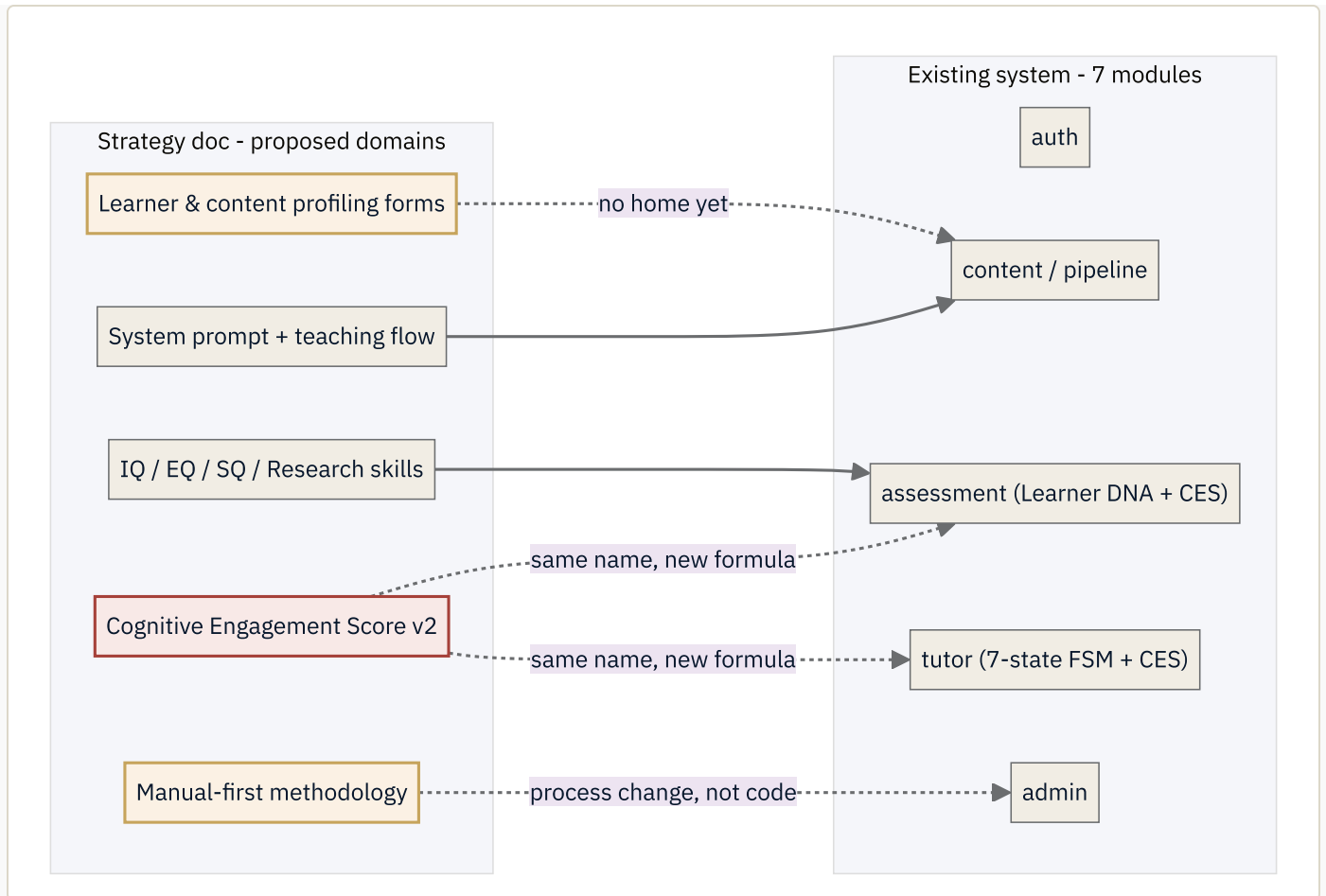
Source plan: AI_Learning_Product_Final_Strategy.pdf

Build snapshot: master-tracker.md, ces-decisions-developer-guide.md **As of:** 2026-09-11

Three things to settle before writing any story

1. **"CES" means two different things.** The live formula gating tutor interventions and the plan's skill-growth score share a name and nothing else. See Domain 5.
2. **SQ and Research have no home in Learner DNA.** The 9 shipped dimensions cover IQ and EQ reasonably well; nothing today measures explanation-to-others or evidence-checking. See Domain 4.
3. **Book- and chapter-level intent isn't captured anywhere.** Today's only inputs are a 20-question diagnostic and a duration pick. Three new forms is real, unscoped engineering. See Domain 1.

- Aligned — built and matches the plan
- Partial — exists, different shape
- Net-new — no code today
- Conflict — needs a decision first



Where each strategy-doc domain lands on the existing module map. Dotted edges mark domains with no current owner or a name that already means something else.

01 **Learner & Content Profiling**

○ net-new

CURRENT	PLANNED
A 20-question diagnostic (OnboardingFlow.tsx , Story 2-3) scores the 9-dimension Learner DNA profile once, at signup. A 3-card Learner Mode picker (Deep / Balanced / Refresher, Stories S2-07–09) sets duration at upload time. Neither captures a stated goal, a deadline, a preferred tone or pace, or anything about <i>why</i> a specific book or chapter was uploaded.	Three layered forms — Onboarding (goal, level, tone, pace), Book (purpose, scope, deadline), Chapter (doubts, focus, expected outcome) — merged into a priority-ordered system prompt, each optional after the first ask.

No schema, storage, or UI exists for book- or chapter-level intent. This is the single largest genuinely new engineering surface in the plan — everything else below extends something already built.

02 Lesson Generation Pipeline

exists, shape undecided

CURRENT

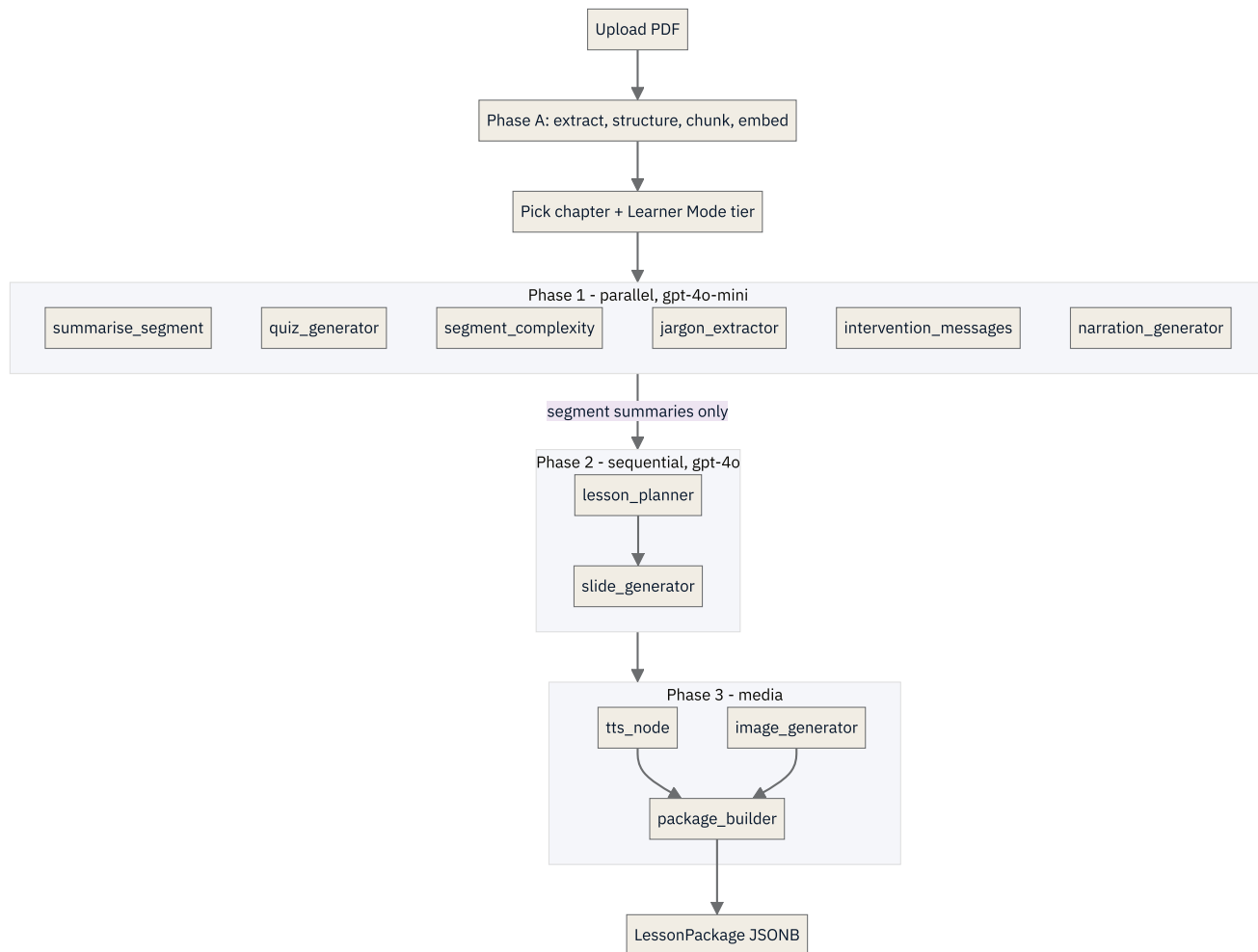
11-node LangGraph pipeline, code-verified end to end. Phase 1 (six economy nodes, `gpt-4o-mini`) runs in parallel; Phase 2 (`lesson_planner`, `slide_generator`, `gpt-4o`) runs sequentially on Phase-1 summaries only, never raw text — a hard 5× token-savings rule.

Two systemic bugs (reducer-field duplication, TTS-fallback script loss) were found and fixed 2026-07-28/29.

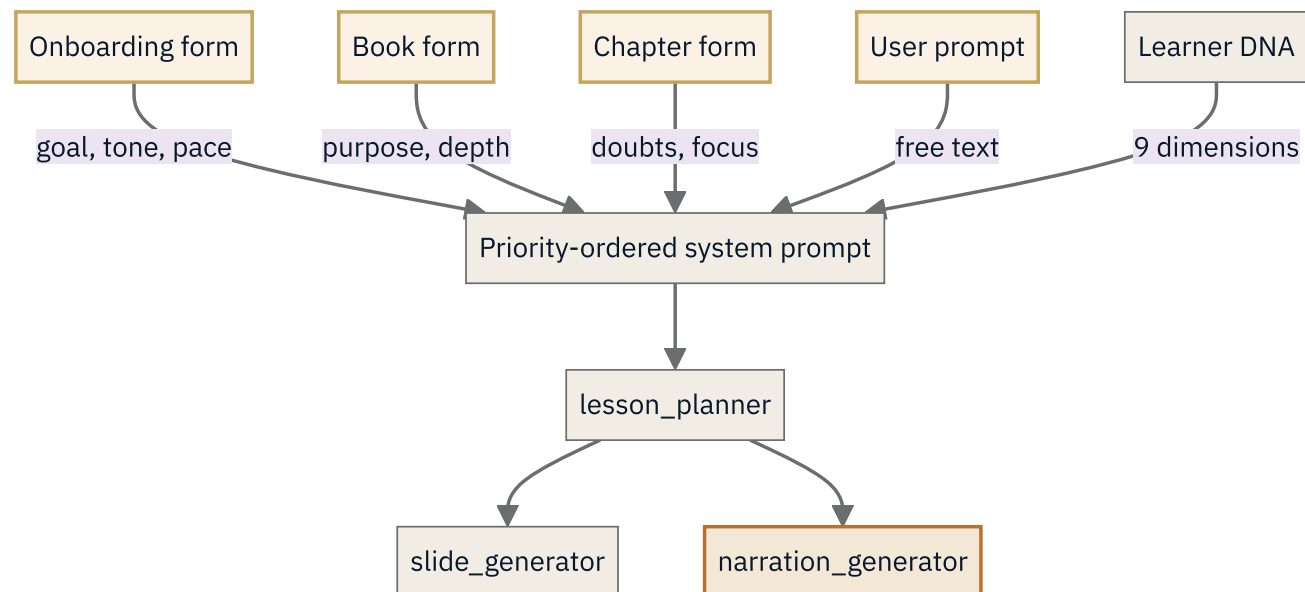
PLANNED

A specific 9-step teaching flow (Introduce → Explain → Simplify → Apply → Ask → Reflect → Correct → Connect → Reinforce) and a 60/40 technical-to-engagement narration ratio. Duration should reshape depth and pacing, not just section count.

The nodes exist; none of their prompts encode this structure yet. Dev 1's own backlog already carries an open, unscoped item — **"redesign lesson_planner/slide_generator prompts: plain-language explanations + easy diagrams"** — that this plan would give real content to.



The pipeline as it runs today. The Phase 1 → Phase 2 handoff is the one architectural rule the plan must not violate: *lesson_planner* never sees raw chapter text.



What the plan would insert upstream of the existing nodes. The 4 new inputs (gold outline) need forms and storage; narration_generator (amber) needs new prompt content, not new code.

03 **Media Generation**

● already aligned

CURRENT	PLANNED
TTS: Sarvam Bulbul v3 → Azure → Browser. Image: Gemini Nano Banana → GPT Image 2 → text-only. Both fallback chains locked and implemented. A human-recorded- narration node is already an open Dev 1 backlog item, for unrelated reasons.	No new media technology — only a content requirement (slides stay sparse, narration carries the teaching).

No action needed here beyond feeding better scripts (Domain 02) into what already exists.

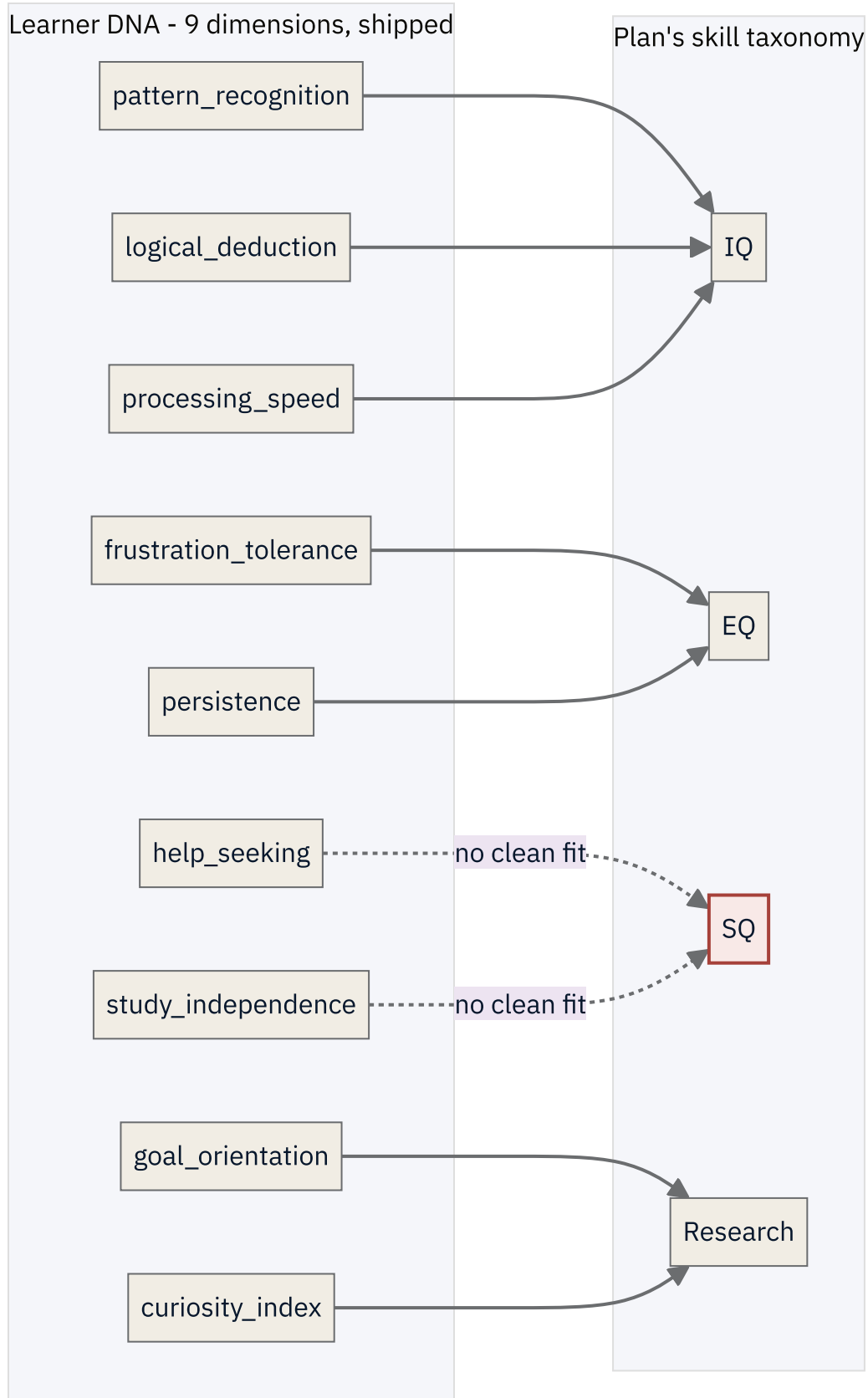
04 Skills Model — Learner DNA vs IQ/EQ/SQ/Research

taxonomy mismatch

CURRENT	PLANNED
9 EMA-fused dimensions in <code>dna_fusion.py</code> : <code>pattern_recognition,</code> <code>logical_deduction,</code> <code>processing_speed,</code> <code>frustration_tolerance,</code> <code>persistence,</code> <code>help_seeking,</code> <code>goal_orientation,</code> <code>curiosity_index,</code> <code>study_independence.</code> Never surfaced as raw scores.	A 4-pillar taxonomy — IQ, EQ, SQ, Research — each with named activities (teach-back for SQ, evidence-checking for Research).

IQ and EQ map cleanly to 3 dimensions each. **SQ and Research don't.**

`help_seeking` / `study_independence` are the closest existing signals to SQ but don't measure explaining a concept to someone else, and nothing today scores source-checking or evidence quality.



Six of nine shipped dimensions map onto IQ or EQ without much argument. SQ (outlined red) is the pillar with no supporting signal today.

05 **Cognitive Engagement Score (CES)**

× naming collision

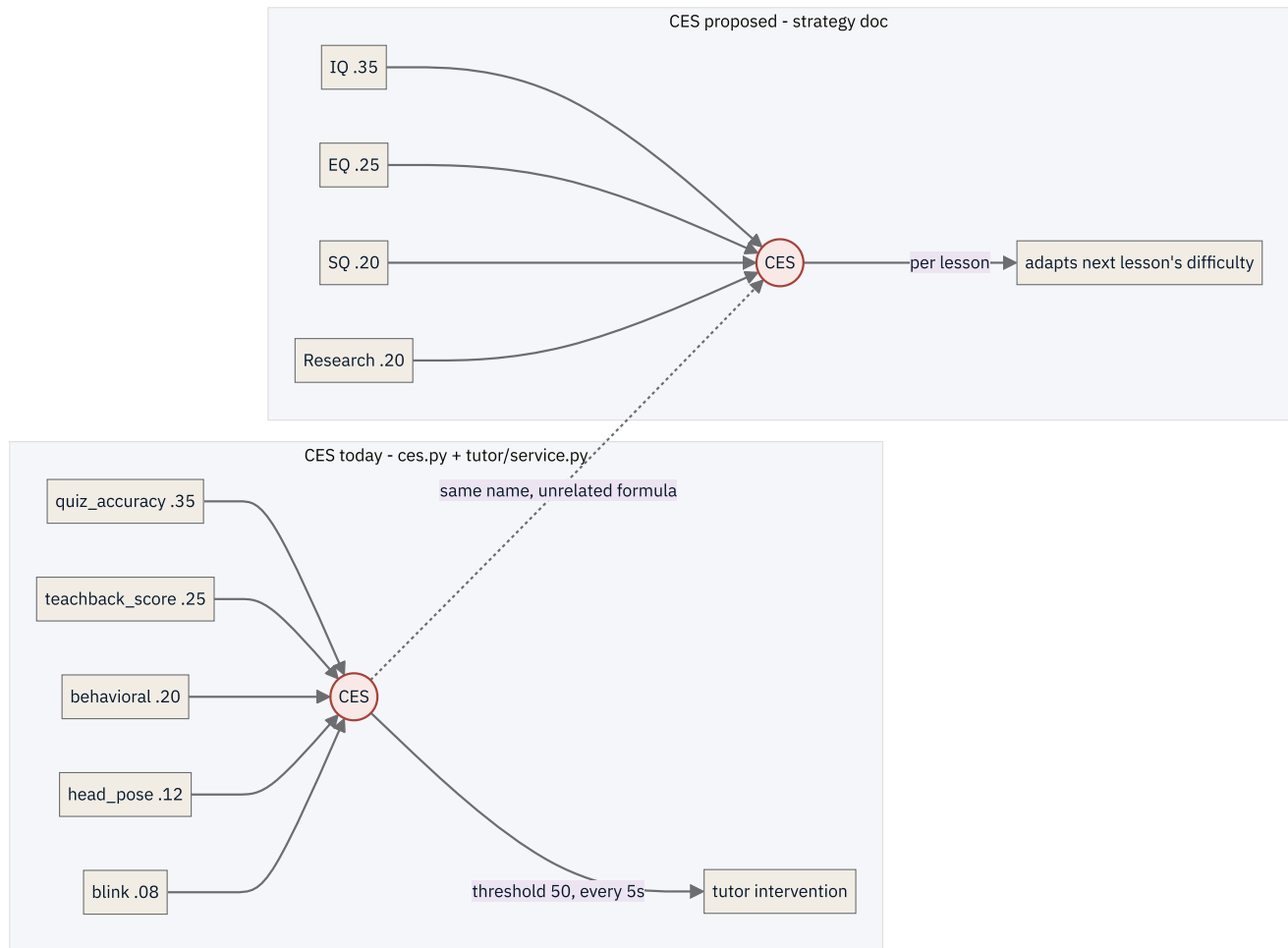
CURRENT

$\text{quiz_accuracy} \times .35 + \text{teachback} \times .25 + \text{behavioral} \times .20 + \text{head_pose} \times .12 + \text{blink} \times .08$. Recomputed every 5s from webcam + quiz/teach-back signals. Gates tutor interventions: 2 consecutive windows below 50 fires a nudge. Weights are env-var-validated to sum to 1.0 at startup.

PLANNED

$\text{IQ} \times .35 + \text{EQ} \times .25 + \text{SQ} \times .20 + \text{Research} \times .20$. A slower, post-lesson skill-growth composite meant to adapt future lesson difficulty. No attention or webcam component at all.

Two unrelated formulas share one name across two teams' mental models. **Resolve before any story references "CES."** Suggestion: keep the name for the live, load-bearing score already gating the tutor FSM; give the plan's version a different name (e.g. "Skill Growth Index") or fold it into Learner DNA's existing growth-tracking instead.



Same three-letter acronym, two formulas with no shared inputs and two different consumers — one real-time and attention-driven, one a slow skill composite.

06

Manual-First Development Methodology

○ process, not code

CURRENT	PLANNED
BMAD story-first process: a story is committed before implementation, and every PR passes a 6-agent adversarial review. That rigor is applied to code the pipeline already generates automatically — never to a hand-built reference lesson.	Hand-build one complete 45-minute lesson first, score it against a Quality Threshold checklist, then treat it as the benchmark the automated pipeline's output is graded against.

This doesn't need a module — it needs a decision to spend one cycle producing a manual reference lesson (using the same underlying models directly) *before* the next round of Domain 02's prompt changes ships.

07

Lesson Formats & Duration

● container exists, contract doesn't

CURRENT	PLANNED
Learner Mode tiers already ship end-to-end: T1 Deep (45m), T2 Balanced (30m), T3 Refresher (15m) — selection screen, tier disclaimers, and tier-to-generation wiring all live (Stories S2-07–09). One open item asks to show explicit "15/30/45 min" labels — cosmetic only.	Duration must change depth, pace, example density, and skill-activity count — not just which tier's pipeline runs or how much gets truncated.

Whether `lesson_planner` actually varies content depth per tier, versus truncating the same plan, hasn't been given the file:line audit treatment the CES formula got. Worth the same scrutiny before assuming it's handled.

08

Real-time Tutor State Machine

●

no functional change

CURRENT	PLANNED
7-state FSM, live, fully tested (884-line suite), gated by the live CES above, with a 2-minute cooldown and a 3-intervention cap per session.	Not addressed directly — but every story that touches this domain inherits Domain 05's naming collision the moment it says "CES."

Annotate, don't rebuild: this FSM is unaffected except by name-collision spillover from Domain 05.

Recap

DOMAIN	STATUS	WHAT IT NEEDS
01 · Profiling forms	○ Net-new	Schema + UI for book/chapter intent; onboarding goal capture beyond the diagnostic.
02 · Pipeline prompts	◐ Partial	Encode the 9-step flow and 60/40 ratio into existing node prompts.
03 · Media generation	● Aligned	Nothing — already matches.
04 · Skills taxonomy	◐ Mismatch	Decide how (or whether) SQ and Research get real signals.
05 · CES	✕ Conflict	Rename one formula before either is referenced in a new story.
06 · Manual-first process	○ Net-new	A sequencing decision, not a build task.
07 · Duration formats	◐ Partial	Verify tier actually changes depth, not just duration label.
08 · Tutor FSM	● Aligned	None — downstream of Domain 05's fix.

Built from AI_Learning_Product_Final_Strategy.pdf against docs/master-tracker.md and docs/ces-decisions-developer-guide.md, transformED repo, 2026-09-11.